BLOCKCHAIN ENGINEERING

ASSIGNMENT - 002

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Batch: 05

Title: Implementation of a Simple Storage Smart Contract

Code -

simple_storage_smart_contract.py.py > SimpleStorage > set

```python
from tkinter import *
import hashlib
import time

# ------------------------------------------------
# SMART CONTRACT SIMULATION
# ------------------------------------------------
class SimpleStorage:
    def __init__(self):
        self.stored_value = 0
        self.transactions = []

    def set(self, value):
        timestamp = time.strftime("%Y-%m-%d %H:%M:%S")
        tx_hash = self.generate_hash(value, timestamp)

        self.stored_value = value

        transaction = {
            "value": value,
            "hash": tx_hash,
            "time": timestamp
        }

```

```python

        self.transactions.append(transaction)
        return transaction

    def get(self):
        return self.stored_value

    def generate_hash(self, value, timestamp):
        data = f"{value}{timestamp}"
        return hashlib.sha256(data.encode()).hexdigest()


# ------------------------------------------------
# CONTRACT OBJECT (DEPLOYMENT)
# ------------------------------------------------
contract = SimpleStorage()

# ------------------------------------------------
# GUI FUNCTIONS
# ------------------------------------------------
def store_value():
    try:
```

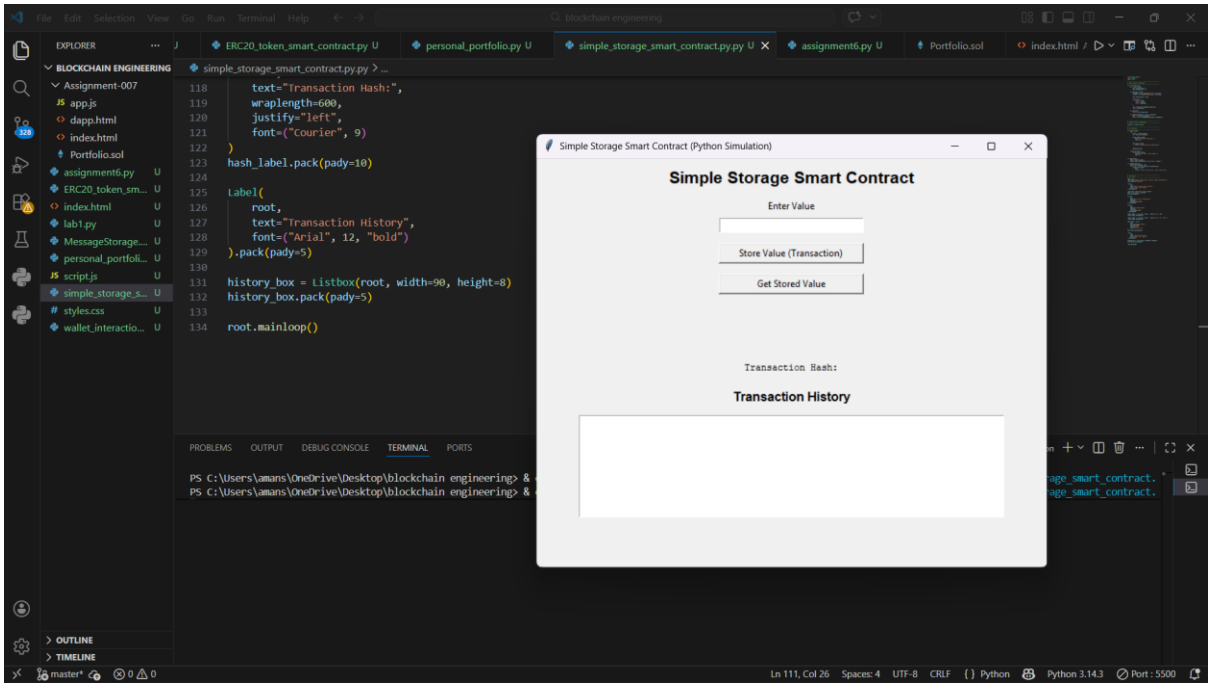
```python
        value = int(entry.get())
        tx = contract.set(value)

        status_label.config(
            text="Transaction Successful ✓",
            fg="green"
        )

        hash_label.config(
            text=f"Transaction Hash:\n{tx['hash']}"
        )

        update_history()

    except ValueError:
        status_label.config(
            text="Please enter a valid number ✗",
            fg="red"
        )

def get_value():
    value = contract.get()
    result_label.config(text=f"Stored Value: {value}")
```

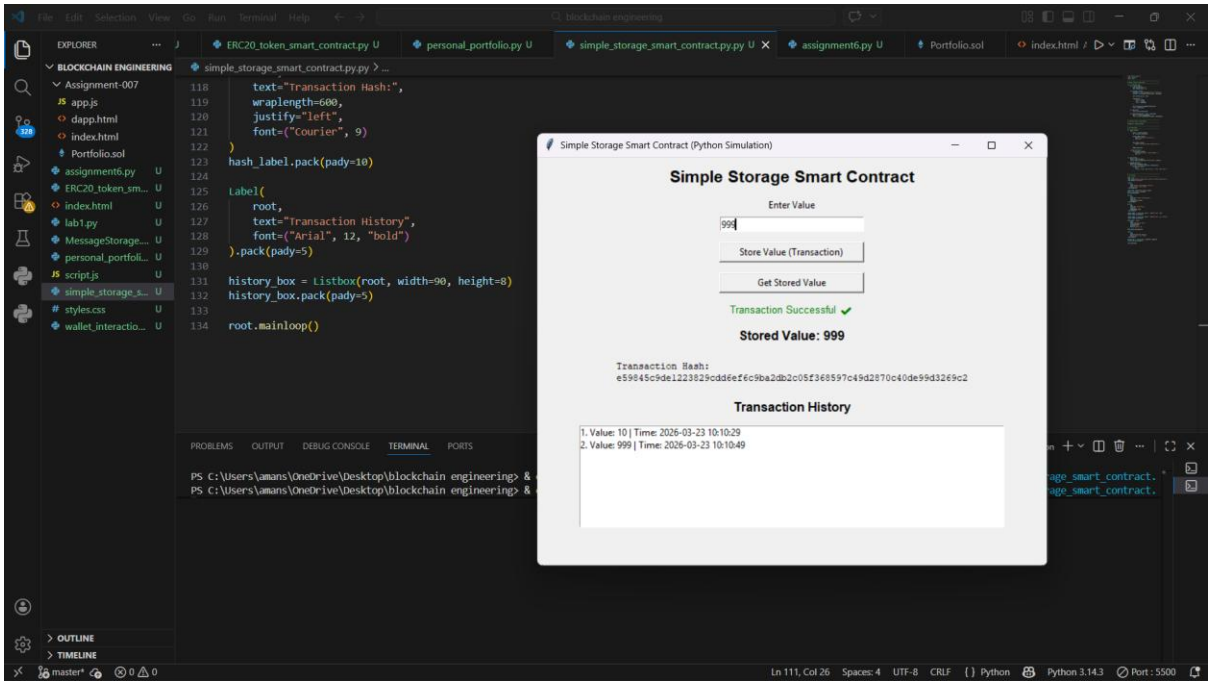
```python
def update_history():
    history_box.delete(0, END)
    for i, tx in enumerate(contract.transactions):
        history_box.insert(
            END,
            f"{i+1}. Value: {tx['value']} | Time: {tx['time']}"
        )


# ------------------------------------------------
# GUI DESIGN
# ------------------------------------------------
root = Tk()
root.title("Simple Storage Smart Contract (Python Simulation)")
root.geometry("650x520")

Label(
    root,
    text="Simple Storage Smart Contract",
    font=("Arial", 16, "bold")
).pack(pady=10)

Label(root, text="Enter Value").pack()
entry = Entry(root, width=30)
```

```python
entry.pack(pady=5)

Button(
    root,
    text="Store Value (Transaction)",
    width=25,
    command=store_value
).pack(pady=8)

Button(
    root,
    text="Get Stored Value",
    width=25,
    command=get_value
).pack(pady=5)

status_label = Label(root, text="", font=("Arial", 10))
status_label.pack(pady=5)

result_label = Label(root, text="", font=("Arial", 12, "bold"))
result_label.pack(pady=5)

hash_label = Label(
    root,
```

```python
        text="Transaction Hash:",
        wraplength=600,
        justify="left",
        font=("Courier", 9)
    )
    hash_label.pack(pady=10)

    Label(
        root,
        text="Transaction History",
        font=("Arial", 12, "bold")
    ).pack(pady=5)

    history_box = Listbox(root, width=90, height=8)
    history_box.pack(pady=5)

    root.mainloop()
```

Output -




simple_storage_smart_contract.py.py
text="Transaction Hash:",
wraplength=600,
justify="left",
font=("Courier", 9)
)
hash_label.pack(pady=10)
Label(
root,
text="Transaction History",
font=("Arial", 12, "bold")
).pack(pady=5)
history_box = Listbox(root, width=90, height=8)
history_box.pack(pady=5)
root.mainloop()
Simple Storage Smart Contract (Python Simulation)
Simple Storage Smart Contract
Enter Value
999
Store Value (Transaction)
Get Stored Value
Transaction Successful
Stored Value: 999
Transaction Hash:
e59845c9de1223829cdd6ef6c9ba2db2c05f368597c49d2870c40de99d3269c2
Transaction History
1. Value: 10 | Time: 2026-03-23 10:10:29
2. Value: 999 | Time: 2026-03-23 10:10:49